

Accounting Business and Society

Week 10

Tutorial A

Exercise 1

Background:

This activity will allow you to reinforce your understanding on the decentralised organisation structure and the existence of different responsibility centres within the decentralised organisations.

You will be working in a pair for this activity.

Task:

1. You and your peer are required to discuss about the differences between decentralisation and centralisation as the common organisation structures.
2. While you need to find and explain 3 advantages of decentralisation, your peer will need to find and explain 3 advantages of centralisation.
3. You and your peer are required to discuss and write everything you may know about the following chart and how it is relevant to the decentralised organisation structure. The following reading, What is responsibility centre, provided below can help you with some relevant information to complete this task.



Exercise 2



Background:

As a continuation from the previous activity, this one will provide you with an opportunity to understand in depth of the revenue centre , one of four key responsibility centres in an organisation.

Task:

You are required to find necessary information from the site, Revenue Centre: Complete Guide provided below to complete the following tasks:

1. How does Revenue centre fit into an organisation structure?
2. What are common authority of Revenue Centre Manager?
3. Can you provide 2 examples of Revenue centres in the practice?
4. How is the performance of Revenue centre measured?
5. What might be the drawbacks of using Revenue Centres to measure performance?

Resources:

Revenue Centre: Complete Guide

Exercise 3

Background:

The following activity will provide you an opportunity to practically apply the evaluation tools including ROI and Residual Income to measure the performance of different investment units.

Task:

1. You are required to access to the ROI vs Residual Income Excel Spreadsheet provided below.
2. Conduct a performance evaluation of 3 different investment segments for Linh Do Accounting Firm for the year 2022 using Margin, Turnover, ROI and Residual Income.

Resources:

ROI vs Residual Income Excel Spreadsheet

Exercise 4



PROBLEM 11–14 Measures of Internal Business Process Performance LO11–3
DataSpan, Inc., automated its plant at the start of the current year and installed a flexible manufacturing system. The company is also evaluating its suppliers and moving toward Lean Production. Many adjustment problems have been encountered, including problems relating to performance measurement. After much study, the company has decided to use the performance measures below, and it has gathered data relating to these measures for the first four months of operations.

	Month			
	1	2	3	4
Throughput time (days)	?	?	?	?
Delivery cycle time (days)	?	?	?	?
Manufacturing cycle efficiency (MCE)	?	?	?	?
Percentage of on-time deliveries	91%	86%	83%	79%
Total sales (units)	3,210	3,072	2,915	2,806

Exercise 4



Management has asked for your help in computing throughput time, delivery cycle time, and MCE. The following average times have been logged over the last four months:

	<u>Average per Month (in days)</u>			
	1	2	3	4
Move time per unit	0.4	0.3	0.4	0.4
Process time per unit	2.1	2.0	1.9	1.8
Wait time per order before start of production	16.0	17.5	19.0	20.5
Queue time per unit	4.3	5.0	5.8	6.7
Inspection time per unit	0.6	0.7	0.7	0.6

Exercise 4



Required:

1. For each month, compute the following:
 - a. The throughput time.
 - b. The delivery cycle time.
 - c. The manufacturing cycle efficiency (MCE).
2. Evaluate the company's performance over the last four months.
3. Refer to the move time, process time, and so forth, given above for month 4.
 - a. Assume that in month 5 the move time, process time, and so forth, are the same as in month 4, except that through the use of Lean Production the company is able to completely eliminate the queue time during production. Compute the new throughput time and MCE.
 - b. Assume in month 6 that the move time, process time, and so forth, are again the same as in month 4, except that the company is able to completely eliminate both the queue time during production and the inspection time. Compute the new throughput time and MCE.

Exercise 5

PROBLEM 11–15 Return on Investment (ROI) and Residual Income LO11–1, LO11–2



Financial data for Joel de Paris, Inc., for last year follow:

Joel de Paris, Inc. Balance Sheet

	Beginning Balance	Ending Balance
Assets		
Cash	\$ 140,000	\$ 120,000
Accounts receivable	450,000	530,000
Inventory	320,000	380,000
Plant and equipment, net	680,000	620,000
Investment in Buisson, S.A.	250,000	280,000
Land (undeveloped)	180,000	170,000
Total assets	<u>\$2,020,000</u>	<u>\$2,100,000</u>
Liabilities and Stockholders' Equity		
Accounts payable	\$ 360,000	\$ 310,000
Long-term debt	1,500,000	1,500,000
Stockholders' equity	160,000	290,000
Total liabilities and stockholders' equity	<u>\$2,020,000</u>	<u>\$2,100,000</u>

Exercise 5



Joel de Paris, Inc. Income Statement

Sales		\$4,050,000
Operating expenses		<u>3,645,000</u>
Net operating income		405,000
Interest and taxes:		
Interest expense	\$150,000	
Tax expense	<u>110,000</u>	<u>260,000</u>
Net income		<u><u>\$ 145,000</u></u>

The company paid dividends of \$15,000 last year. The “Investment in Buisson, S.A.,” on the balance sheet represents an investment in the stock of another company. The company’s minimum required rate of return of 15%.

Required:

1. Compute the company’s average operating assets for last year.
2. Compute the company’s margin, turnover, and return on investment (ROI) for last year. (Hint: Should you use net income or net operating income in your calculations?)
3. What was the company’s residual income last year?

Exercise 6



PROBLEM 11–17 Comparison of Performance Using Return on Investment (ROI) LO11–1

Comparative data on three companies in the same service industry are given below:

	Company		
	A	B	C
Sales	\$600,000	\$500,000	\$?
Net operating income	\$ 84,000	\$ 70,000	\$?
Average operating assets	\$300,000	\$?	\$1,000,000
Margin	?	?	3.5%
Turnover	?	?	2
ROI	?	7%	?

Required:

1. What advantages are there to breaking down the ROI computation into two separate elements, margin and turnover?
2. Fill in the missing information above, and comment on the relative performance of the three companies in as much detail as the data permit. Make specific recommendations about how to improve the ROI.

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Week 10

Tutorial B

Exercise 1

Background:

The students should have attempted the Segmented Income Statement Excel Spreadsheet before turning up the tutorial B this week.

Task:

1. You now will be working in a pair with your mate to discuss and compare your solution on your Segmented Income Statement Excel Spreadsheet (10 mins)
2. After discussion time, your pair might be requested to present the solution to the class.

Resources:

Segmented Income Statement

Exercise 2



Understanding Balanced scorecard

Background:

This activity will give you an opportunity to enhance your understanding on Balanced Scorecard and how it is practically used by the company, Rockwater which is a worldwide leader in underwater engineering and construction, to develop the strategic planning framework and measure the firm's performance. You will be working in a group of 4 people for this activity.

Task:

You are required to access the following Harvard Business Review site, Putting the Balanced Scorecard to work by Robert S. Kaplan and David P. Norton, to complete the following tasks.

1. What is the basic background of Rockwater Company?
2. What are strategic goals developed from four different perspectives of Balanced scorecard?
3. Each member will need to stand in one perspective of BSC (Financial, Customer, Internal Business Process, Learning & Growth) to explain to the group ONE(1) key measure used by Rockwater.

Resources:

Putting the Balanced Scorecard to Work

Exercise 3



PROBLEM 11–18 Return on Investment (ROI) and Residual Income LO11–1, LO11–2

“I know headquarters wants us to add that new product line,” said Dell Havasi, manager of Billings Company’s Office Products Division. “But I want to see the numbers before I make any move. Our division’s return on investment (ROI) has led the company for three years, and I don’t want any letdown.”

Billings Company is a decentralized wholesaler with five autonomous divisions. The divisions are evaluated on the basis of ROI, with year-end bonuses given to the divisional managers who have the highest ROIs. Operating results for the company’s Office Products Division for this year are given below:

Sales	\$10,000,000
Variable expenses	<u>6,000,000</u>
Contribution margin	4,000,000
Fixed expenses	<u>3,200,000</u>
Net operating income	<u><u>\$ 800,000</u></u>
Divisional average operating assets	<u><u>\$4,000,000</u></u>

Exercise 3



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The company had an overall return on investment (ROI) of 15% this year (considering all divisions). Next year the Office Products Division has an opportunity to add a new product line that would require an additional investment that would increase average operating assets by \$1,000,000. The cost and revenue characteristics of the new product line per year would be:

Sales	\$2,000,000
Variable expenses	60% of sales
Fixed expenses	\$640,000

Required:

1. Compute the Office Products Division's ROI for this year.
2. Compute the Office Products Division's ROI for the new product line by itself.
3. Compute the Office Products Division's ROI for next year assuming that it performs the same as this year and adds the new product line.
4. If you were in Dell Havasi's position, would you accept or reject the new product line? Explain.
5. Why do you suppose headquarters is anxious for the Office Products Division to add the new product line?

Exercise 3



6. Suppose that the company's minimum required rate of return on operating assets is 12% and that performance is evaluated using residual income.
 - a. Compute the Office Products Division's residual income for this year.
 - b. Compute the Office Products Division's residual income for the new product line by itself.
 - c. Compute the Office Products Division's residual income for next year assuming that it performs the same as this year and adds the new product line.
 - d. Using the residual income approach, if you were in Dell Havasi's position, would you accept or reject the new product line? Explain.

Exercise 4

PROBLEM 11–19 Internal Business Process Performance Measures LO11–3

Tombro Industries is in the process of automating one of its plants and developing a flexible manufacturing system. The company is finding it necessary to make many changes in operating procedures. Progress has been slow, particularly in trying to develop new performance measures for the factory. In an effort to evaluate performance and determine where improvements can be made, management has gathered the following data relating to activities over the last four months:

	Month			
	1	2	3	4
Quality control measures:				
Number of defects	185	163	124	91
Number of warranty claims	46	39	30	27
Number of customer complaints	102	96	79	58
Material control measures:				
Purchase order lead time	8 days	7 days	5 days	4 days
Scrap as a percent of total cost	1%	1%	2%	3%
Machine performance measures:				
Machine downtime as a percentage of availability	3%	4%	4%	6%
Use as a percentage of availability	95%	92%	89%	85%
Setup time (hours)	8	10	11	12
Delivery performance measures:				
Throughput time	?	?	?	?
Manufacturing cycle efficiency (MCE)	?	?	?	?
Delivery cycle time	?	?	?	?
Percentage of on-time deliveries	96%	95%	92%	89%

Exercise 4

The president has read in industry journals that throughput time, MCE, and delivery cycle time are important measures of performance, but no one is sure how they are computed. You have been asked to assist the company, and you have gathered the following data relating to these measures:

	Average per Month (in days)			
	1	2	3	4
Wait time per order before start of production	9.0	11.5	12.0	14.0
Inspection time per unit	0.8	0.7	0.7	0.7
Process time per unit	2.1	2.0	1.9	1.8
Queue time per unit	2.8	4.4	6.0	7.0
Move time per unit	0.3	0.4	0.4	0.5

Exercise 4

Required:

1. For each month, compute the following performance measures:
 - a. Throughput time.
 - b. MCE.
 - c. Delivery cycle time.
2. Using the performance measures given in the main body of the problem and the performance measures computed in (1) above, do the following:
 - a. Identify areas where the company seems to be improving.
 - b. Identify areas where the company seems to be deteriorating.
3. Refer to the inspection time, process time, and so forth, given for month 4.
 - a. Assume that in month 5 the inspection time, process time, and so forth, are the same as for month 4, except that the company is able to completely eliminate the queue time during production using Lean Production. Compute the new throughput time and MCE.
 - b. Assume that in month 6 the inspection time, process time, and so forth, are the same as in month 4, except that the company is able to eliminate both the queue time during production and the inspection time using Lean Production. Compute the new throughput time and MCE.

Exercise 5



PROBLEM 11–20 Return on Investment (ROI) Analysis LO11–1

The contribution format income statement for Huerra Company for last year is given below:

	Total	Unit
Sales	\$4,000,000	\$80.00
Variable expenses	<u>2,800,000</u>	<u>56.00</u>
Contribution margin	1,200,000	24.00
Fixed expenses	<u>840,000</u>	<u>16.80</u>
Net operating income	360,000	7.20
Income taxes @ 30%	<u>108,000</u>	<u>2.16</u>
Net income	<u><u>\$ 252,000</u></u>	<u><u>\$ 5.04</u></u>

The company had average operating assets of \$2,000,000 during the year.

Exercise 5



Required:

1. Compute the company's return on investment (ROI) for the period using the ROI formula stated in terms of margin and turnover.

For each of the following questions, indicate whether the margin and turnover will increase, decrease, or remain unchanged as a result of the events described, and then compute the new ROI figure. Consider each question separately, starting in each case from the data used to compute the original ROI in (1) above.

2. Using Lean Production, the company is able to reduce the average level of inventory by \$400,000. (The released funds are used to pay off short-term creditors.)

3. The company achieves a cost savings of \$32,000 per year by using less costly materials.

4. The company issues bonds and uses the proceeds to purchase machinery and equipment that increases average operating assets by \$500,000. Interest on the bonds is \$60,000 per year. Sales remain unchanged. The new, more efficient equipment reduces production costs by \$20,000 per year.

5. As a result of a more intense effort by salespeople, sales are increased by 20%; operating assets remain unchanged.

6. At the beginning of the year, obsolete inventory carried on the books at a cost of \$40,000 is scrapped and written off as a loss.

7. At the beginning of the year, the company uses \$200,000 of cash (received on accounts receivable) to repurchase and retire some of its common stock.